

eQTL

Jeff Leek

- Dependencies
- Download the data
- eQTL is linear regression
- Fitting many eQTL models with MatrixEQTL
 - Set general parameters
 - Set the data up
 - Running MatrixEQTL
 - Understanding the results
- More information
- Session information

Dependencies

This document depends on the following packages:

```
library(devtools)
library(Biobase)
library(MatrixEQTL)
```

To install these packages you can use the code (or if you are compiling the document, remove the `eval=FALSE` from the chunk.)

```
install.packages(c("devtools", "MatrixEQTL"))
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("Biobase"))
```

Download the data

Here we are going to follow along with the tutorial on MatrixEQTL (http://www.bios.unc.edu/research/genomic_software/Matrix_eQTL/runit.html). First we find the files

```
base.dir = find.package("MatrixEQTL")
SNP_file_name = paste(base.dir, "/data/SNP.txt", sep="");
expression_file_name = paste(base.dir, "/data/GE.txt", sep="")
covariates_file_name = paste(base.dir, "/data/Covariates.txt", sep="")
output_file_name = tempfile()
```

Next we load the data so we can see it

```
expr = read.table(expression_file_name, sep="\t",
                  header=T, row.names=1)
expr[1,]
```

```
##           Sam_01 Sam_02 Sam_03 Sam_04 Sam_05 Sam_06 Sam_07 Sam_08 Sam_09
## Gene_01   4.91   4.63   5.18   5.07   5.74   5.09   5.31   5.29   4.73
##           Sam_10 Sam_11 Sam_12 Sam_13 Sam_14 Sam_15 Sam_16
## Gene_01   5.72   4.75   4.54   5.01   5.03   4.84   4.44
```

```
snps = read.table(SNP_file_name,sep="\t",
                  header=T,row.names=1)
snps[1,]
```

```
##           Sam_01 Sam_02 Sam_03 Sam_04 Sam_05 Sam_06 Sam_07 Sam_08 Sam_09
## Snp_01      2      0      2      0      2      1      2      1      1
##           Sam_10 Sam_11 Sam_12 Sam_13 Sam_14 Sam_15 Sam_16
## Snp_01      1      2      2      1      2      2      1
```

```
cvrt = read.table(covariates_file_name,sep="\t",
                  header=T,row.names=1)
```

eQTL is linear regression

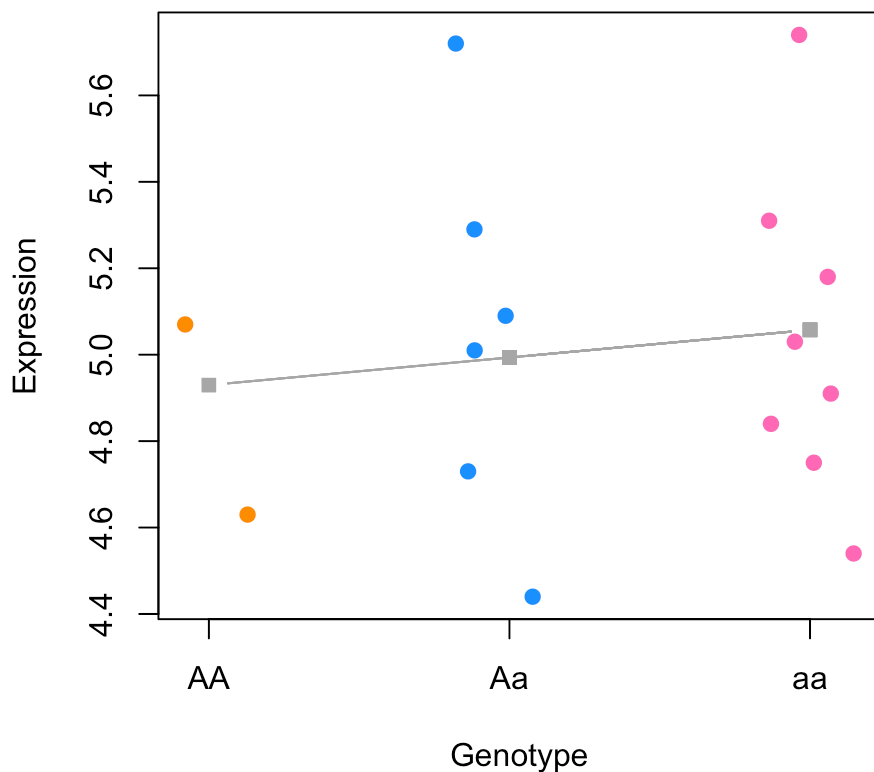
The simplest eQTL analysis just computes linear regression models for each SNP/gene pair.

```
e1 = as.numeric(expr[1,])
s1 = as.numeric(snps[1,])
lm1 = lm(e1 ~ s1)
tidy(lm1)
```

```
##           term   estimate std.error  statistic      p.value
## 1 (Intercept) 4.92967742 0.2137508  23.0627348 1.546033e-12
## 2           s1 0.06387097 0.1386998   0.4604979 6.522304e-01
```

We can visualize the data and the model fits

```
plot(e1 ~ jitter(s1),
     col=(s1+1),xaxt="n",xlab="Genotype",ylab="Expression")
axis(1,at=c(0:2),labels=c("AA","Aa","aa"))
lines(lm1$fitted ~ s1,type="b",pch=15,col="darkgrey")
```



Fitting many eQTL models with MatrixEQTL

Set general parameters

We need to set up the p-value cutoff and the error model (in this case assuming independent errors)

```
pvOutputThreshold = 1e-2
errorCovariance = numeric()
useModel = modelLINEAR
```

Set the data up

Now we need to set up the snp and gene expression data in the special format required by the `MatrixEQTL` package

```
snps = SlicedData$new()
snps$fileDelimiter = "\t"      # the TAB character
snps$fileOmitCharacters = "NA" # denote missing values;
snps$fileSkipRows = 1         # one row of column labels
snps$fileSkipColumns = 1      # one column of row labels
snps$fileSliceSize = 2000     # read file in pieces of 2,000 rows
snps$LoadFile( SNP_file_name )
```

```
## Rows read: 15 done.
```

```
gene = SlicedData$new()  
gene$fileDelimiter = "\t"      # the TAB character  
gene$fileOmitCharacters = "NA" # denote missing values;  
gene$fileSkipRows = 1         # one row of column labels  
gene$fileSkipColumns = 1      # one column of row labels  
gene$fileSliceSize = 2000     # read file in pieces of 2,000 rows  
gene$LoadFile(expression_file_name)
```

```
## Rows read: 10 done.
```

```
cvrt = SlicedData$new()
```

Running MatrixEQTL

We can now run the code to calculate the eQTL that we are interested in

```
me = Matrix_eQTL_engine(  
  snps = snps,  
  gene = gene,  
  cvrt = cvrt,  
  output_file_name = NULL,  
  pvOutputThreshold = pvOutputThreshold,  
  useModel = useModel,  
  errorCovariance = errorCovariance,  
  verbose = TRUE,  
  pvalue.hist = TRUE,  
  min.pv.by.genesnp = FALSE,  
  noFDRsaveMemory = FALSE);
```

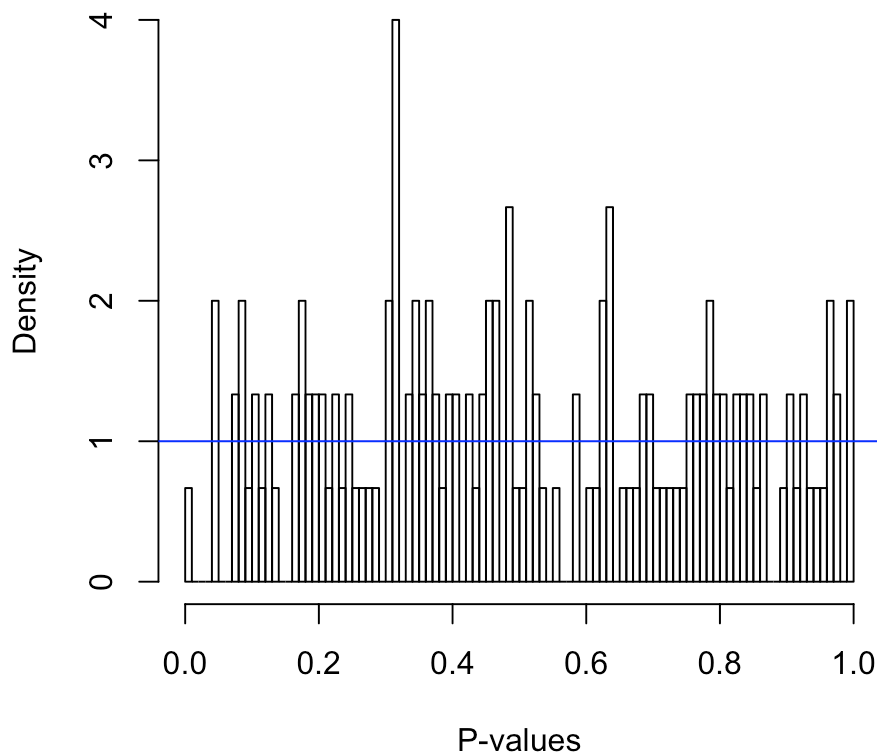
```
## Processing covariates  
## Task finished in 0.001 seconds  
## Processing gene expression data (imputation, residualization, etc.)  
## Task finished in 0.001 seconds  
## Creating output file(s)  
## Task finished in 0.006 seconds  
## Performing eQTL analysis  
## 100.00% done, 1 eQTLs  
## Task finished in 0.01 seconds  
##
```

Understanding the results

We can make a plot of all the p-values from the tests

```
plot(me)
```

Histogram for all 150 p-values



We can also figure look at the number and type of eQTL

```
me$all$neqtls
```

```
## [1] 1
```

```
me$all$eqtls
```

```
##      snps    gene statistic      pvalue FDR      beta
## 1 Snp_11 Gene_06 -3.007106 0.00941791    1 -0.2916667
```

More information

eQTL is an entire field of research.

- MatrixEQTL package (http://www.bios.unc.edu/research/genomic_software/Matrix_eQTL/)
- MatrixEQTL paper (<http://bioinformatics.oxfordjournals.org/content/28/10/1353.abstract?keytype=ref&ijkey=zjMWpHTAUk7OJFw>)
- Surrogate variables applied to eQTL (<http://journals.plos.org/plosgenetics/article?id=10.1371/journal.pgen.0030161>)
- A Bayesian sva framework for eQTL (<http://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1000770>)
- Bioconductor eQTL workflow (<http://www.bioconductor.org/help/workflows/eQTL/>)

Session information

Here is the session information

```
devtools::session_info()
```

```

## setting value
## version R version 3.2.1 (2015-06-18)
## system x86_64, darwin10.8.0
## ui RStudio (0.99.447)
## language (EN)
## collate en_US.UTF-8
## tz America/New_York
##
## package * version date
## acepack 1.3-3.3 2014-11-24
## annotate 1.46.1 2015-07-11
## AnnotationDbi * 1.30.1 2015-04-26
## assertthat 0.1 2013-12-06
## BiasedUrn * 1.06.1 2013-12-29
## Biobase * 2.28.0 2015-04-17
## BiocGenerics * 0.14.0 2015-04-17
## BiocInstaller * 1.18.4 2015-07-22
## BiocParallel 1.2.20 2015-08-07
## biomaRt 2.24.0 2015-04-17
## Biostrings 2.36.3 2015-08-12
## bitops 1.0-6 2013-08-17
## bladderbatch * 1.6.0 2015-08-26
## broom * 0.3.7 2015-05-06
## caTools 1.17.1 2014-09-10
## cluster 2.0.3 2015-07-21
## colorspace 1.2-6 2015-03-11
## corpcor 1.6.8 2015-07-08
## curl 0.9.2 2015-08-08
## DBI * 0.3.1 2014-09-24
## dendextend * 1.1.0 2015-07-31
## DESeq2 * 1.8.1 2015-05-02
## devtools * 1.8.0 2015-05-09
## digest 0.6.8 2014-12-31
## dplyr * 0.4.3 2015-09-01
## edge * 2.1.0 2015-09-06
## evaluate 0.7.2 2015-08-13
## foreign 0.8-66 2015-08-19
## formatR 1.2 2015-04-21
## Formula * 1.2-1 2015-04-07
## futile.logger 1.4.1 2015-04-20
## futile.options 1.0.0 2010-04-06
## gdata 2.17.0 2015-07-04
## genefilter * 1.50.0 2015-04-17
## geneLenDataBase * 1.4.0 2015-09-06
## geneplotter 1.46.0 2015-04-17
## GenomeInfoDb * 1.4.2 2015-08-15
## GenomicAlignments 1.4.1 2015-04-24
## GenomicFeatures 1.20.2 2015-08-14
## GenomicRanges * 1.20.5 2015-06-09
## genstats * 0.1.02 2015-09-05
## ggplot2 * 1.0.1 2015-03-17
## git2r 0.11.0 2015-08-12

```

##	GO.db	3.1.2	2015-09-06
##	goseq	* 1.20.0	2015-04-17
##	gplots	* 2.17.0	2015-05-02
##	gridExtra	2.0.0	2015-07-14
##	gtable	0.1.2	2012-12-05
##	gtools	3.5.0	2015-05-29
##	highr	0.5	2015-04-21
##	HistData	* 0.7-5	2014-04-26
##	Hmisc	* 3.16-0	2015-04-30
##	htmltools	0.2.6	2014-09-08
##	httr	1.0.0	2015-06-25
##	IRanges	* 2.2.7	2015-08-09
##	KernSmooth	2.23-15	2015-06-29
##	knitr	* 1.11	2015-08-14
##	lambda.r	1.1.7	2015-03-20
##	lattice	* 0.20-33	2015-07-14
##	latticeExtra	0.6-26	2013-08-15
##	lazyeval	0.1.10	2015-01-02
##	limma	* 3.24.15	2015-08-06
##	lme4	1.1-9	2015-08-20
##	locfit	1.5-9.1	2013-04-20
##	magrittr	1.5	2014-11-22
##	MASS	* 7.3-43	2015-07-16
##	Matrix	* 1.2-2	2015-07-08
##	MatrixEQTL	* 2.1.1	2015-02-03
##	memoise	0.2.1	2014-04-22
##	mgcv	* 1.8-7	2015-07-23
##	minqa	1.2.4	2014-10-09
##	mnormt	1.5-3	2015-05-25
##	munsell	0.4.2	2013-07-11
##	nlme	* 3.1-122	2015-08-19
##	nloptr	1.0.4	2014-08-04
##	nnet	7.3-10	2015-06-29
##	org.Hs.eg.db	* 3.1.2	2015-07-17
##	plyr	1.8.3	2015-06-12
##	preprocessCore	* 1.30.0	2015-04-17
##	proto	0.3-10	2012-12-22
##	psych	1.5.6	2015-07-08
##	qvalue	* 2.0.0	2015-04-17
##	R6	2.1.1	2015-08-19
##	RColorBrewer	1.1-2	2014-12-07
##	Rcpp	* 0.12.0	2015-07-25
##	RcppArmadillo	* 0.5.400.2.0	2015-08-17
##	RCurl	1.95-4.7	2015-06-30
##	reshape2	1.4.1	2014-12-06
##	rmarkdown	0.7	2015-06-13
##	rpart	4.1-10	2015-06-29
##	Rsamtools	1.20.4	2015-06-01
##	RSkittleBrewer	* 1.1	2015-09-05
##	RSQLite	* 1.0.0	2014-10-25
##	rstudioapi	0.3.1	2015-04-07
##	rtracklayer	1.28.9	2015-08-19

##	rversions	1.0.2	2015-07-13
##	S4Vectors	* 0.6.5	2015-09-01
##	scales	0.3.0	2015-08-25
##	snm	1.16.0	2015-04-17
##	snpStats	* 1.18.0	2015-04-17
##	stringi	0.5-5	2015-06-29
##	stringr	1.0.0	2015-04-30
##	survival	* 2.38-3	2015-07-02
##	sva	* 3.14.0	2015-04-17
##	tidyr	0.2.0	2014-12-05
##	UsingR	* 2.0-5	2015-08-06
##	whisker	0.3-2	2013-04-28
##	XML	3.98-1.3	2015-06-30
##	xml2	0.1.2	2015-09-01
##	xtable	1.7-4	2014-09-12
##	XVector	0.8.0	2015-04-17
##	yaml	2.1.13	2014-06-12
##	zlibbioc	1.14.0	2015-04-17
##	source		
##	CRAN (R 3.2.0)		
##	Bioconductor		
##	Bioconductor		
##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.0)		
##	Bioconductor		
##	Bioconductor		
##	Bioconductor		
##	Bioconductor		
##	Bioconductor		
##	Bioconductor		
##	CRAN (R 3.2.0)		
##	Bioconductor		
##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.1)		
##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.1)		
##	CRAN (R 3.2.1)		
##	CRAN (R 3.2.1)		
##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.1)		
##	Bioconductor		
##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.2)		
##	Github (jdstorey/edge@a1947b5)		
##	CRAN (R 3.2.2)		
##	CRAN (R 3.2.1)		
##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.1)		

```
## Bioconductor
## Bioconductor
## Bioconductor
## Bioconductor
## Bioconductor
## Bioconductor
## local
## CRAN (R 3.2.0)
## CRAN (R 3.2.2)
## Bioconductor
## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.1)
## CRAN (R 3.2.2)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## Bioconductor
## CRAN (R 3.2.2)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
```

```
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## Bioconductor
## Github (alyssafranze/RSkittleBrewer@0a96a20)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## Bioconductor
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.2)
## Bioconductor
## Bioconductor
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.2)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.2)
## CRAN (R 3.2.0)
## Bioconductor
## CRAN (R 3.2.0)
## Bioconductor
```

It is also useful to compile the time the document was processed. This document was processed on: 2015-09-06.